GPIE

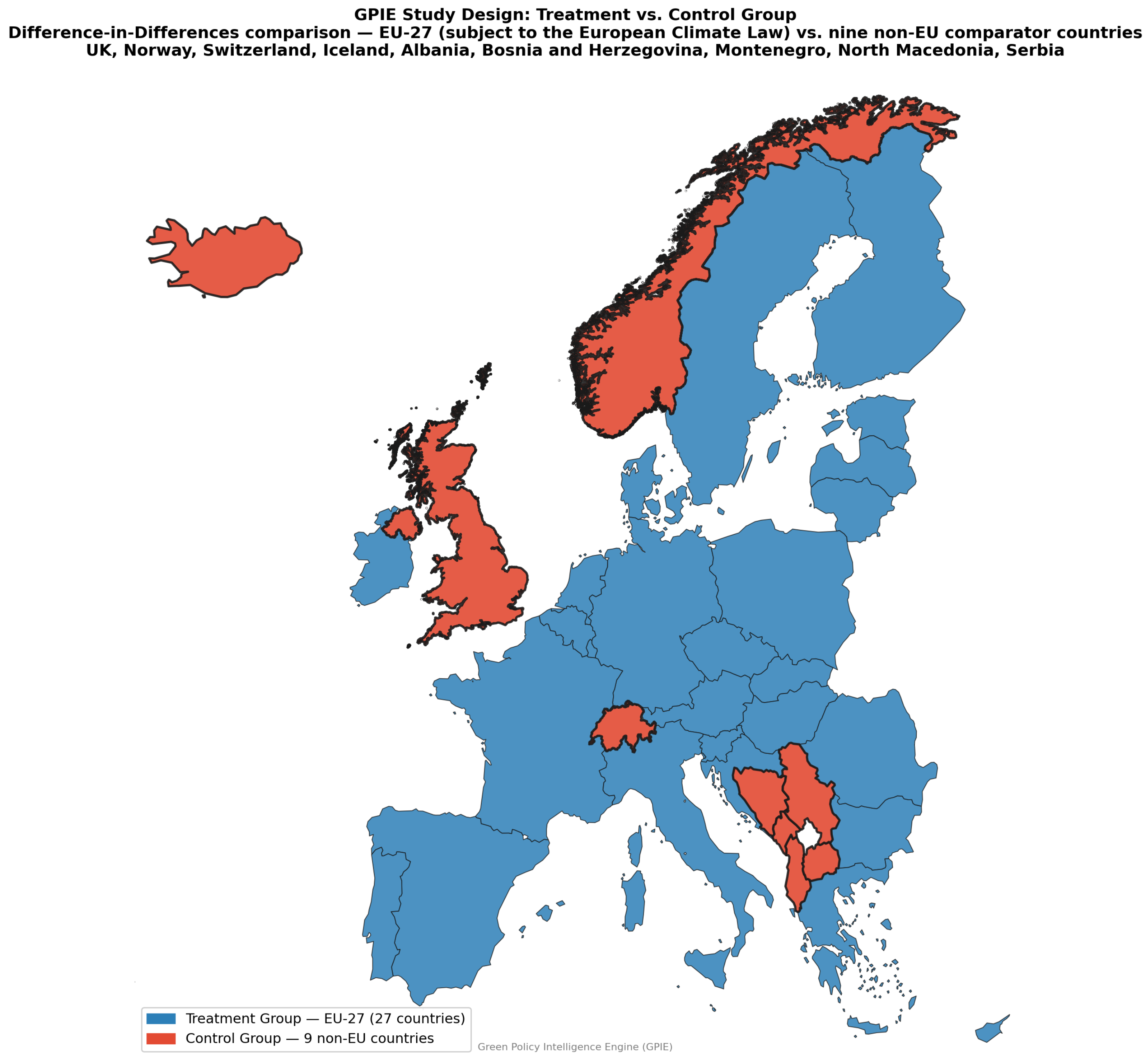
Green Policy Intelligence Engine

All Maps and Plots

18 figures compiled from outputs/plots/

Study Design: Treatment vs. Control Group

EU-27 (treatment) and the 9-country non-EU control group - UK, Norway, Switzerland, Iceland, Albania, Bosnia and Herzegovina, Montenegro, North Macedonia, Serbia.



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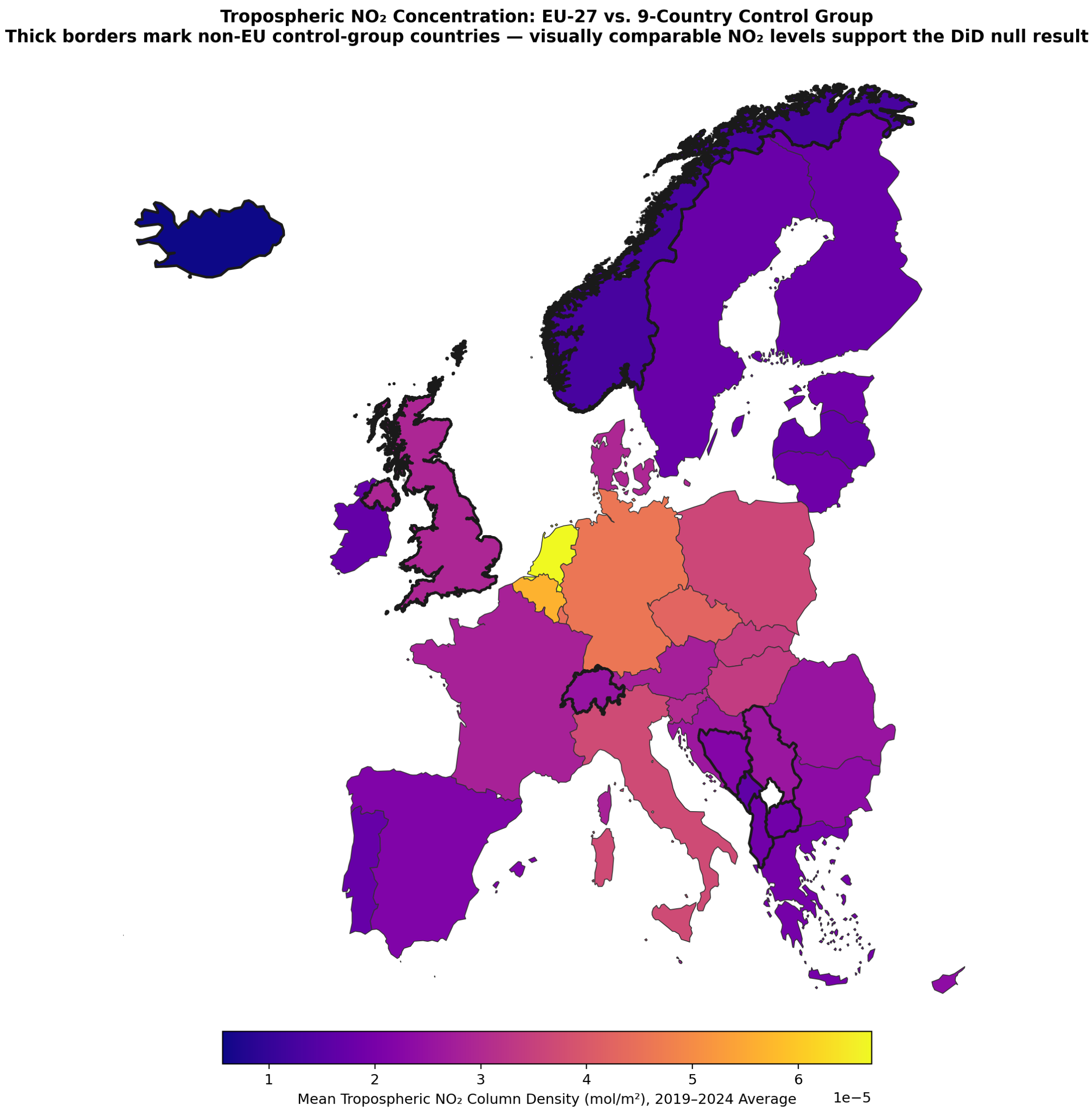
NO2 Choropleth Map

Mean tropospheric NO2 concentration by country, Sentinel-5P TROPOMI.

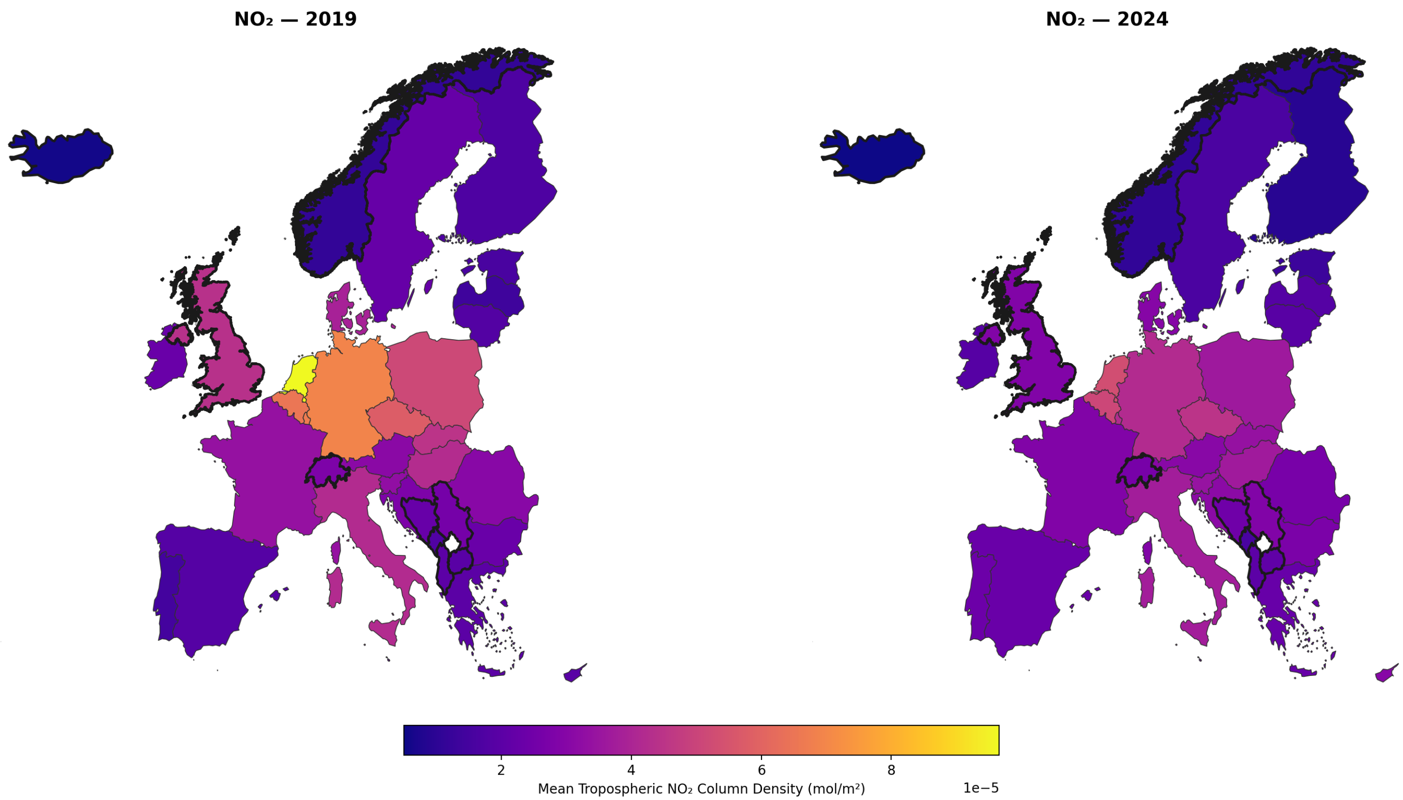


Green Policy Intelligence Engine (GPIE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

NO2 Before vs. After the European Climate Law

Average NO2 across the EU-27, 2019 (pre-treatment) vs. 2024 (post-treatment).

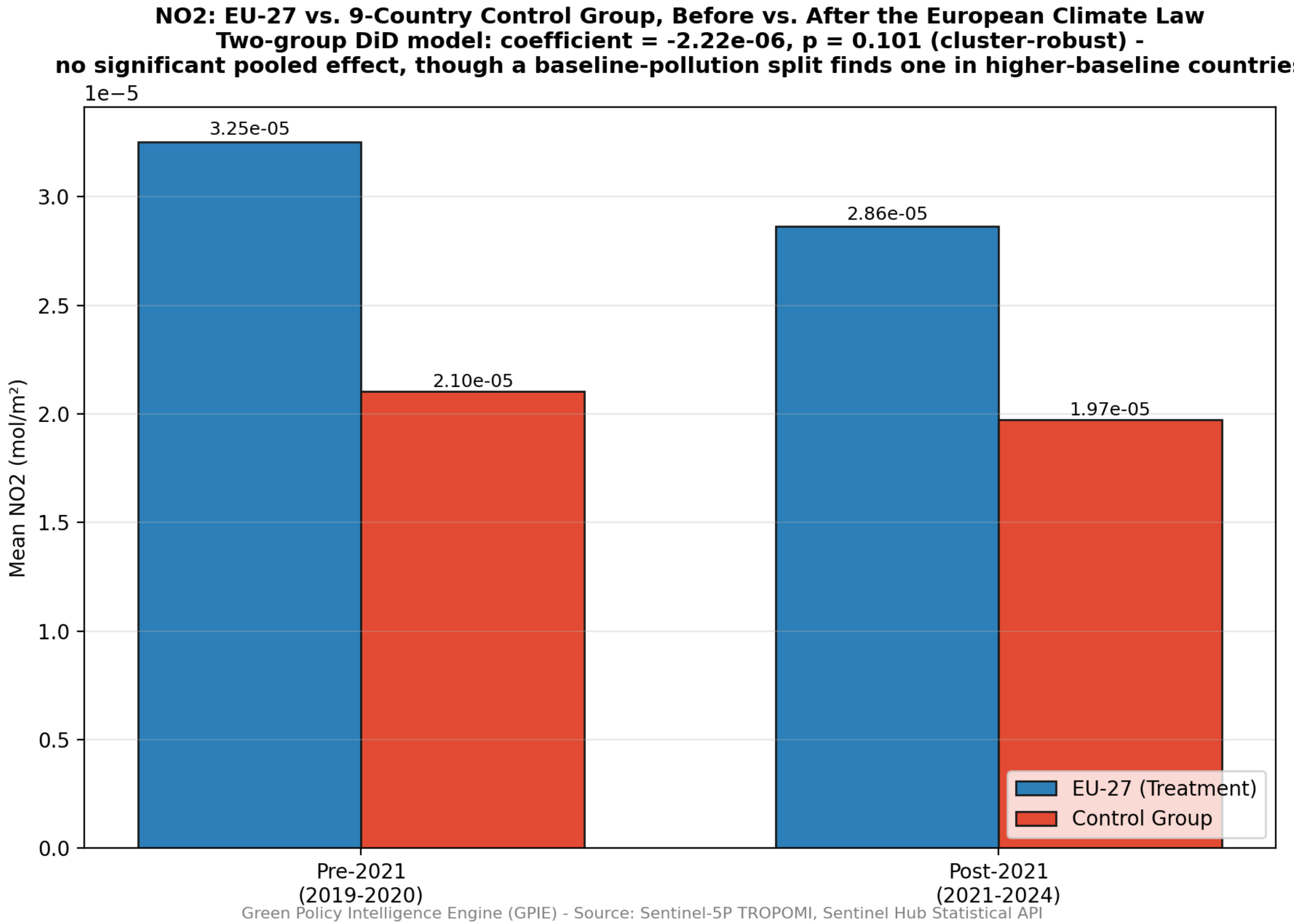
NO_2 Levels Before vs. After the European Climate Law: EU-27 and Control Group (2019 vs. 2024)
Thick borders mark the 9-country non-EU control group — comparable decline pattern across both groups



Green Policy Intelligence Engine (GPIE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

NO2: EU-27 vs. Control Group

Mean NO2, pre- and post-treatment, EU-27 vs. 9-country control group - coefficient = -2.22e-06, p = 0.101.



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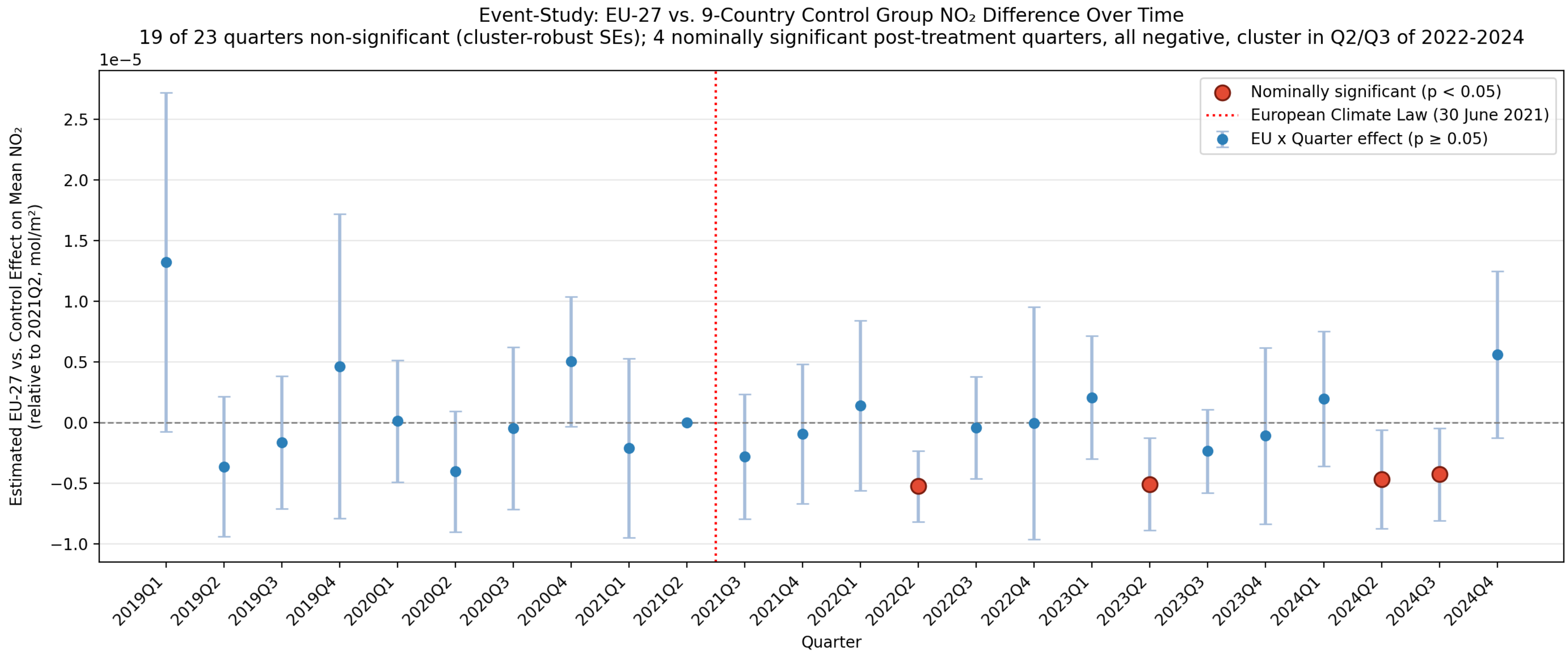
Event-Study: Quarter-by-Quarter NO2 Effect

23-quarter Difference-in-Differences event-study estimates, cluster-robust SEs - 4 significant post-treatment quarters, all negative, clustering in Q2/Q3 of 2022-2024.



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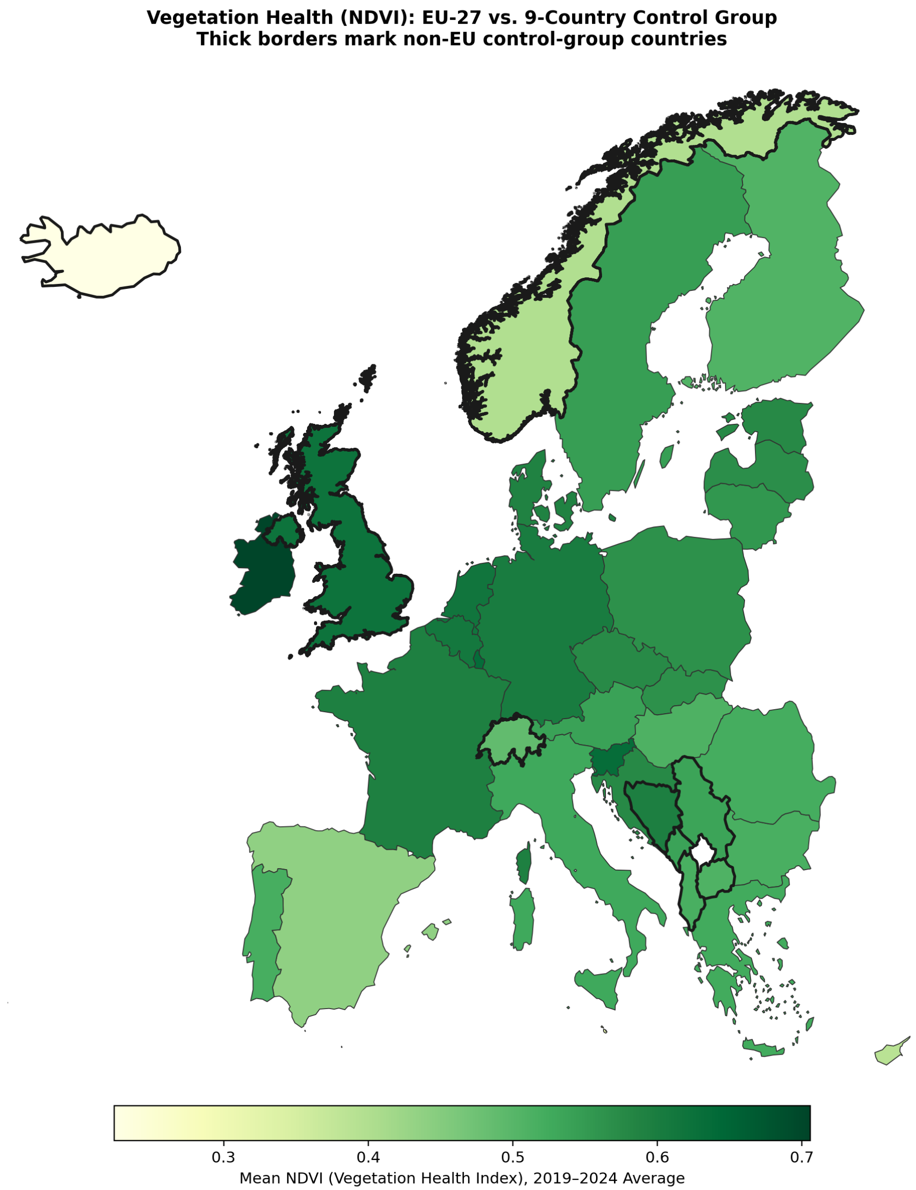
NDVI Choropleth Map

Mean vegetation health index (NDVI) by country.



Green Policy Intelligence Engine (GPIE) — Source: CGLS NDVI 300m, Sentinel Hub Statistical API

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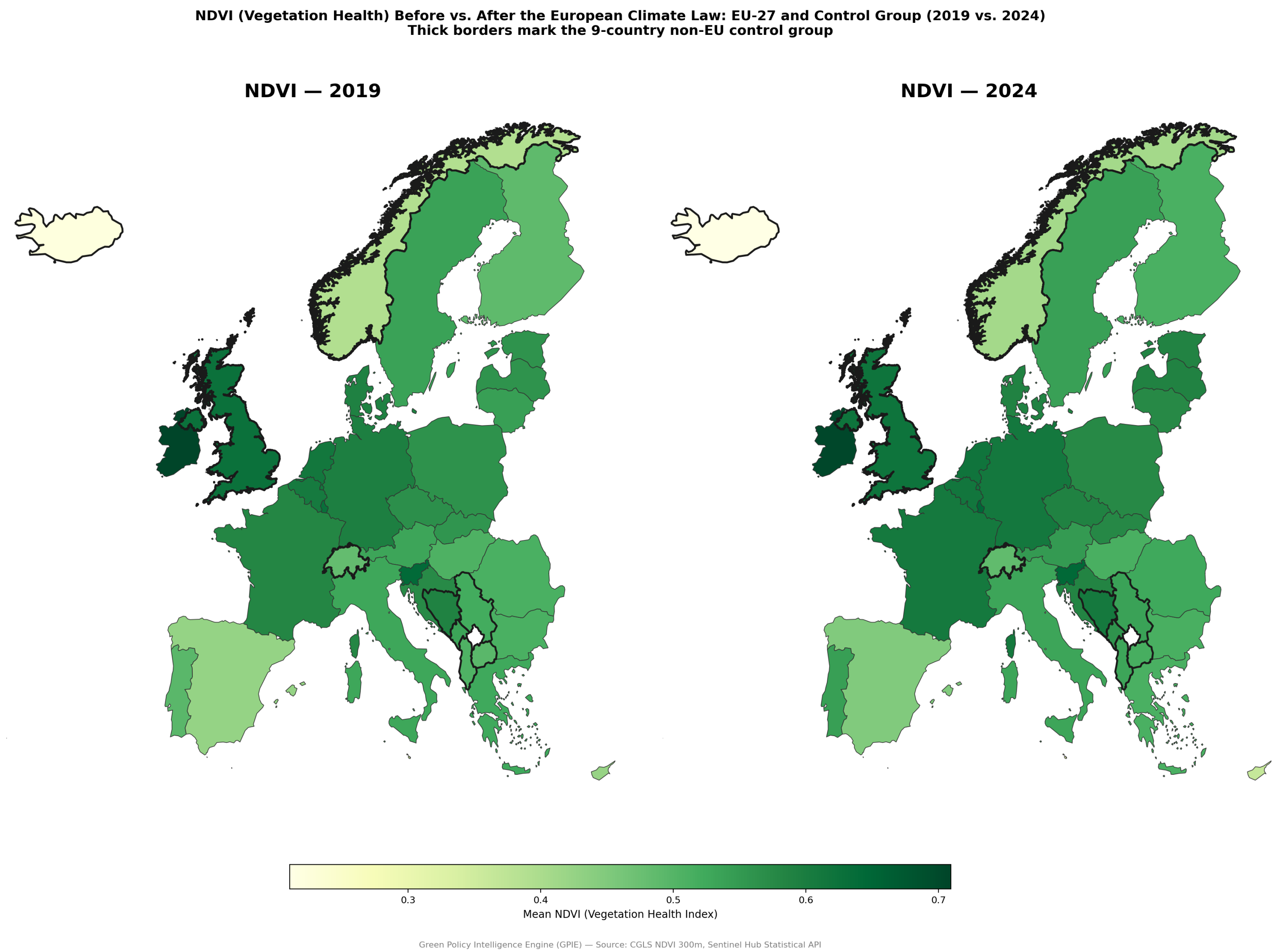
NDVI Before vs. After the European Climate Law

Average NDVI across the EU-27, pre- vs. post-treatment.



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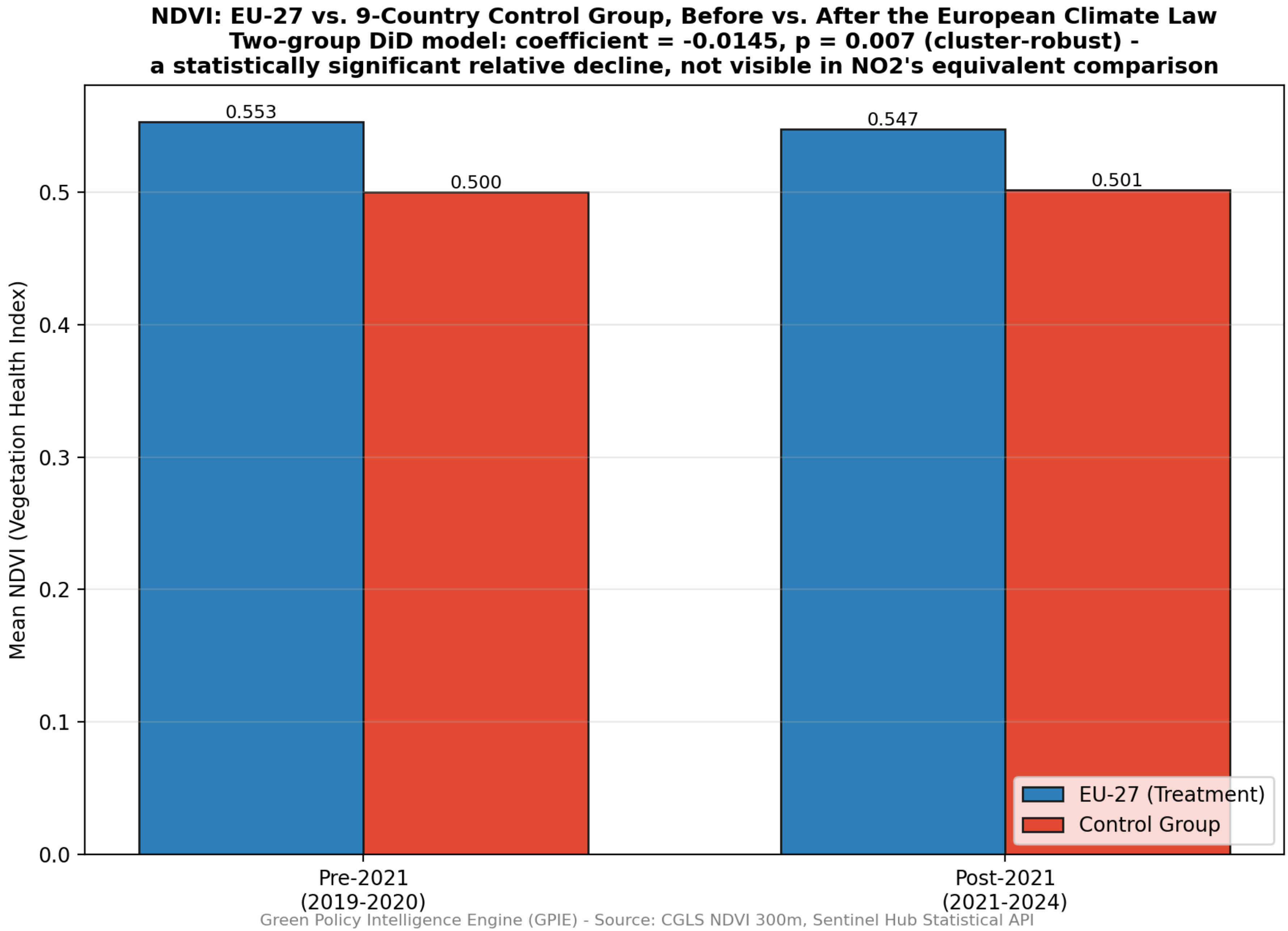
NDVI: EU-27 vs. Control Group

Two-group DiD model: coefficient = -0.0145, p = 0.007 (cluster-robust) - a statistically significant relative decline.



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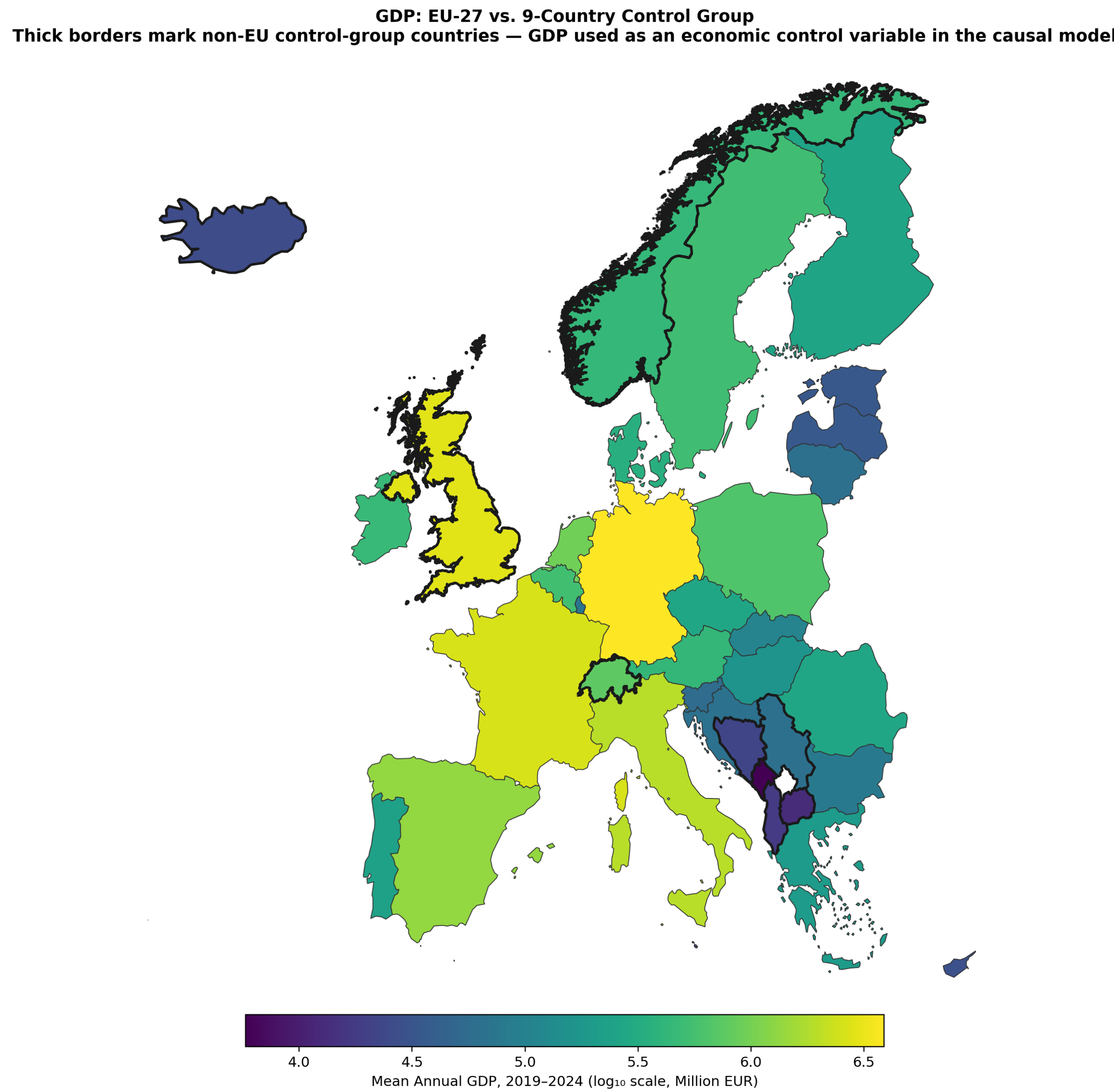
GDP Choropleth Map

Control variable - GDP by country.



Green Policy Intelligence Engine (GPIE) — Source: Eurostat (EU-27), World Bank (control group)

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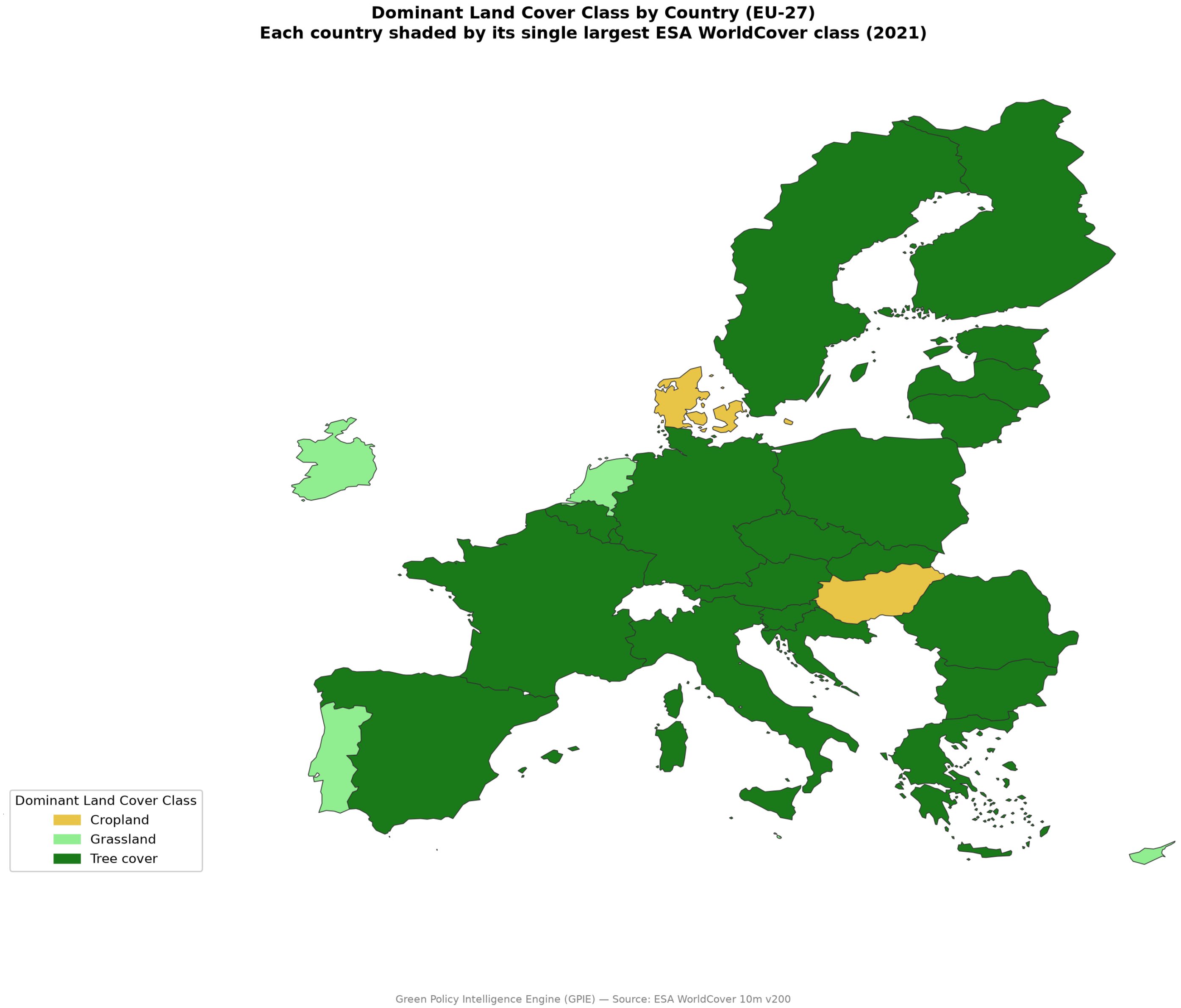
Dominant Land Cover Class Map

Control variable - dominant land-cover classification by country.



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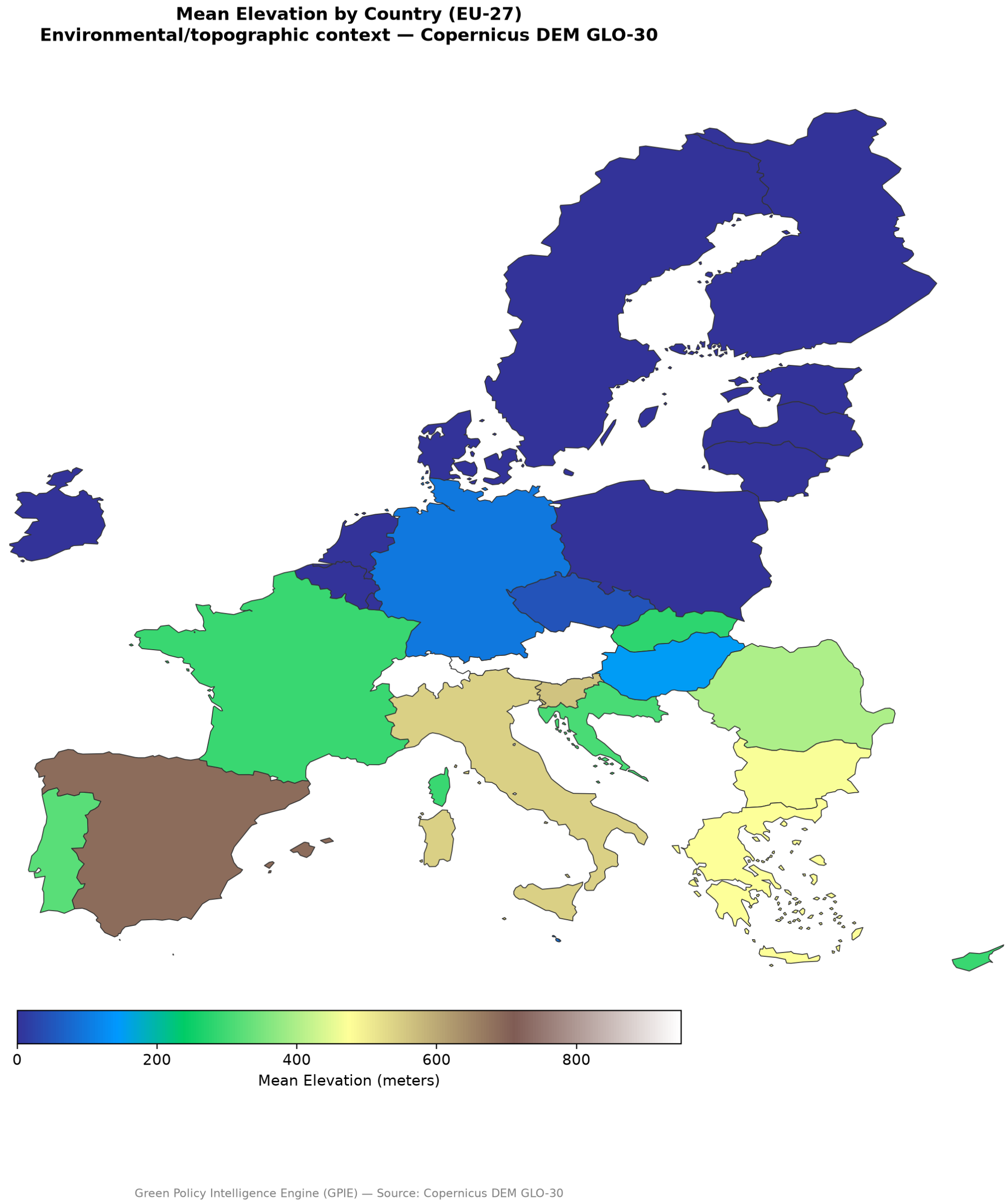
Elevation (DEM) Map

Control variable - digital elevation model.



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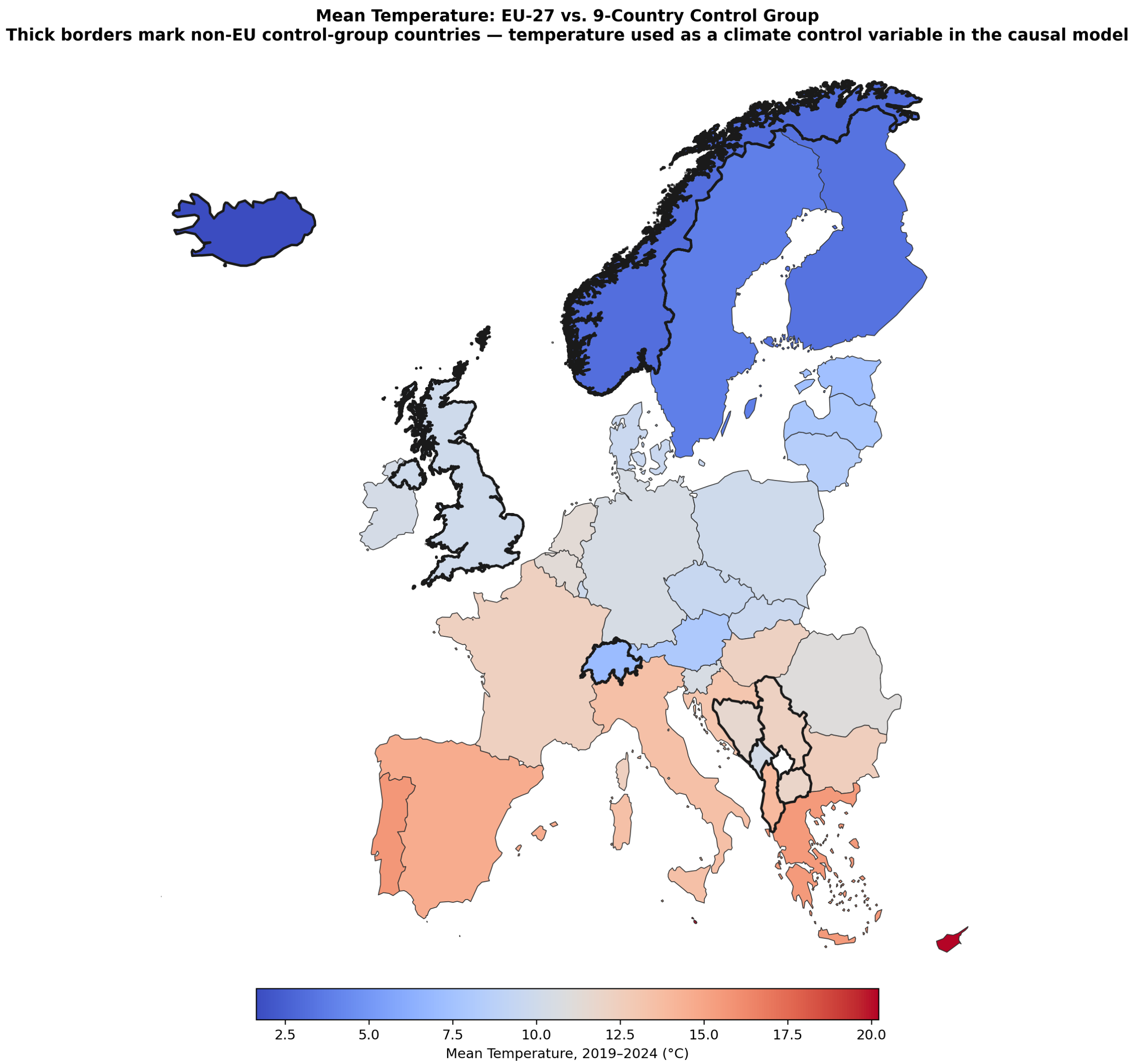
Climate / Temperature Map

Control variable - average temperature by country.



Green Policy Intelligence Engine (GPIE) — Source: ERA5 Reanalysis, Copernicus Climate Data Store

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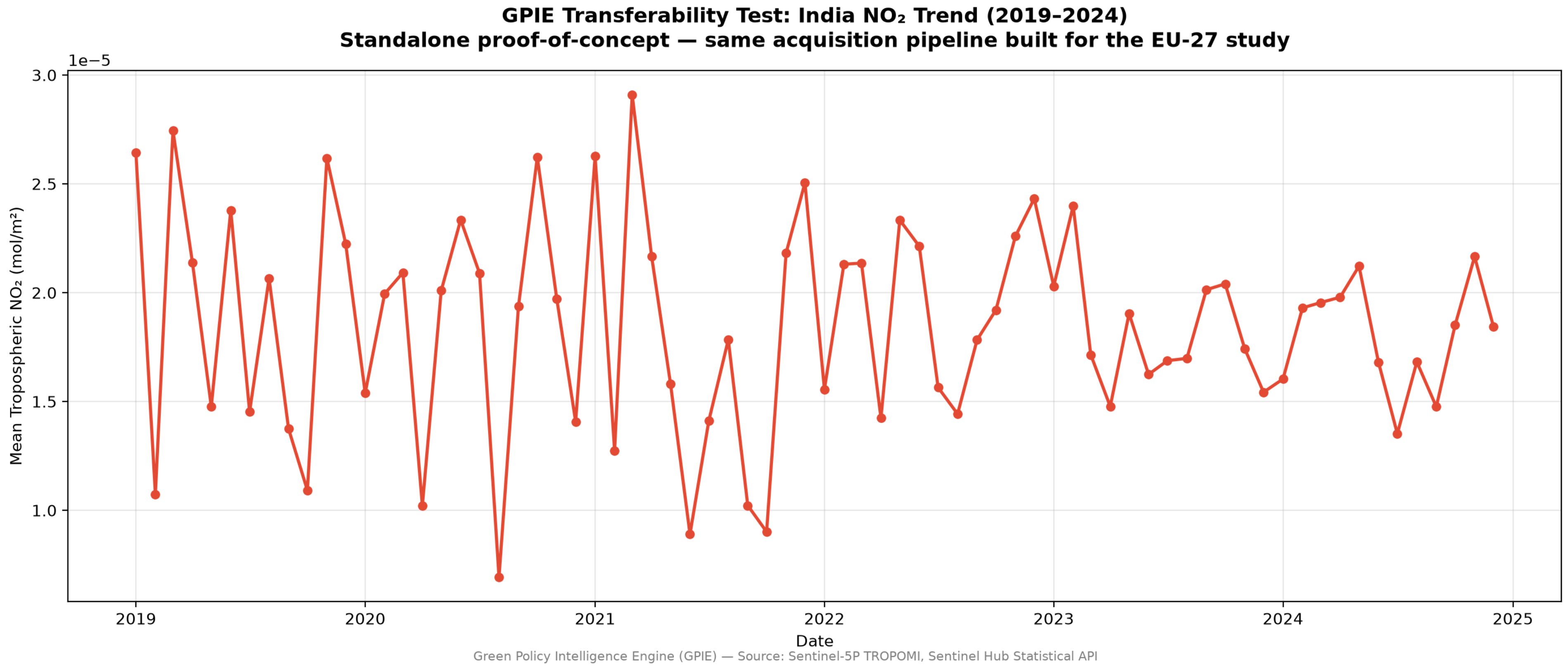
India Transferability Validation

NO2 acquisition pipeline independently tested on India, 2019-2024.



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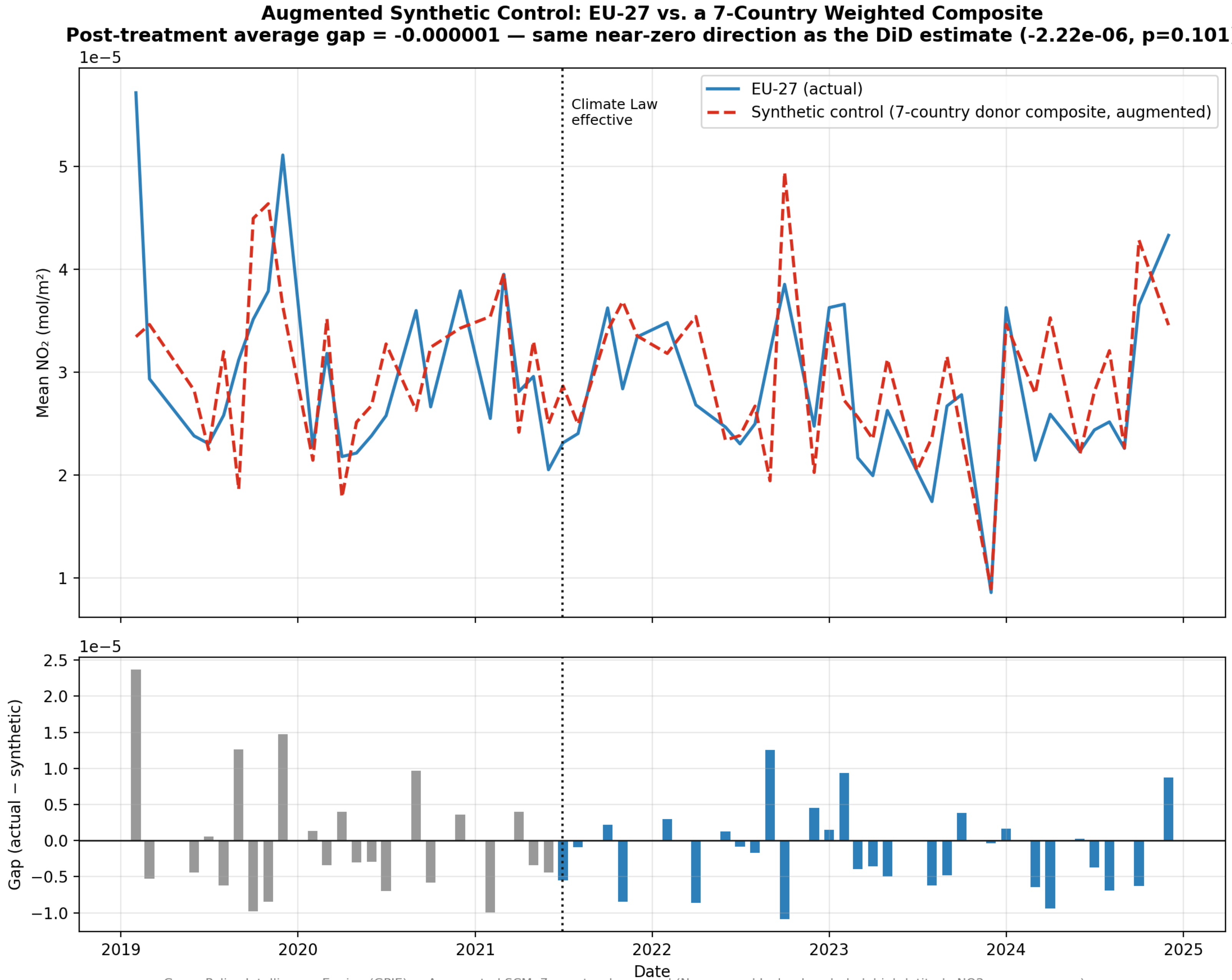
Augmented Synthetic Control: EU-27 vs. 7-Country Donor Composite

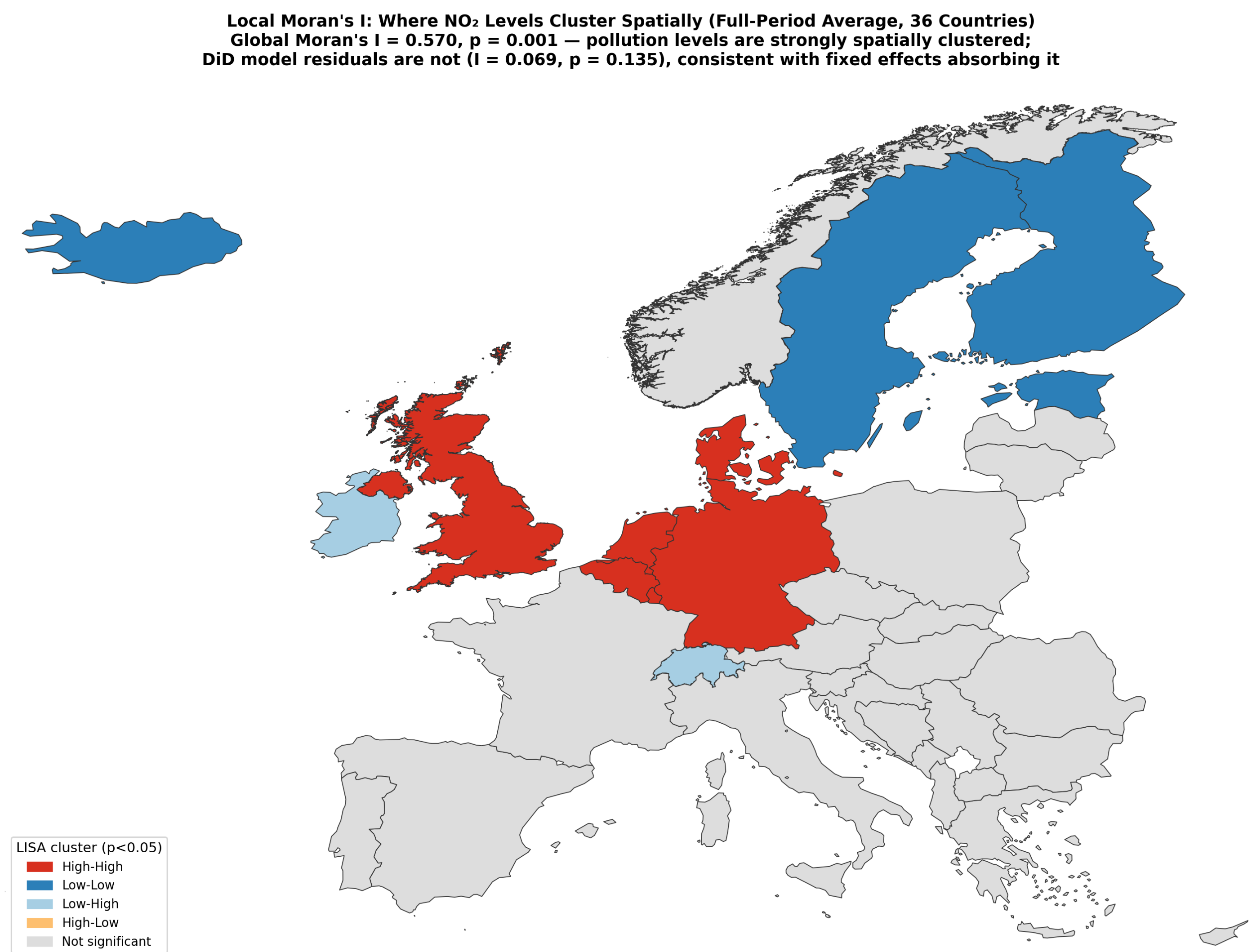
eatment gap = -1e-6, same near-zero direction as the DiD estimate (-2.22e-06, p=0.101), reached independently via a 7-country donor pool (Norway and Iceland excluded, high-latitude NO2 coverage



Green Policy Intelligence Engine (GPIE) — Augmented SCM, 7-country donor pool (Norway and Iceland excluded, high-latitude NO2 coverage gaps)

Local Moran's I (LISA) Spatial Cluster Map

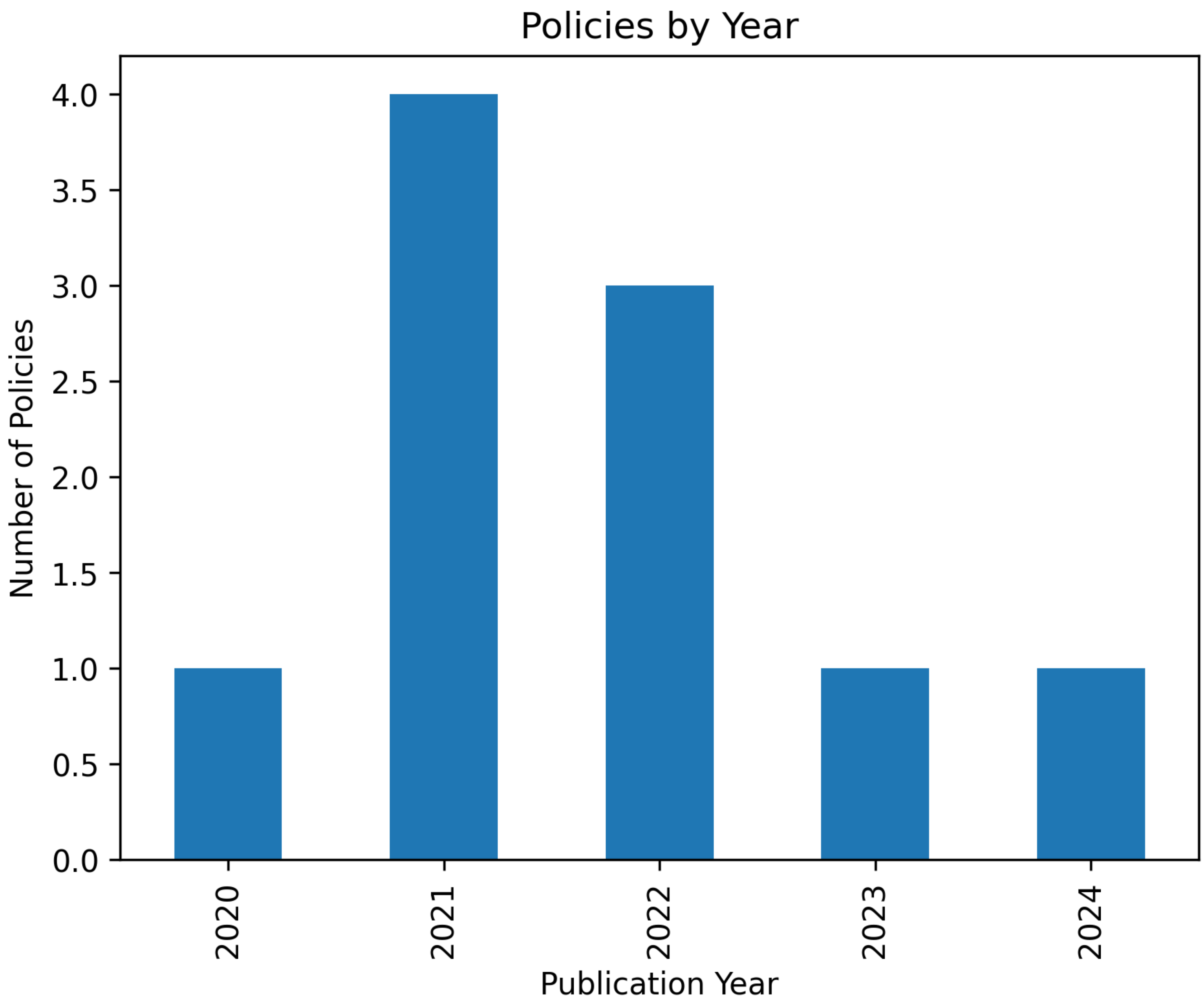
NO2 levels cluster spatially across 36 countries (I=0.570, p=0.001); DiD residuals do not (I=0.069, p=0.135) - fixed effects absorb it.



Green Policy Intelligence Engine (GPIE) — Local Moran's I (esda), KNN-4 spatial weights on country centroids

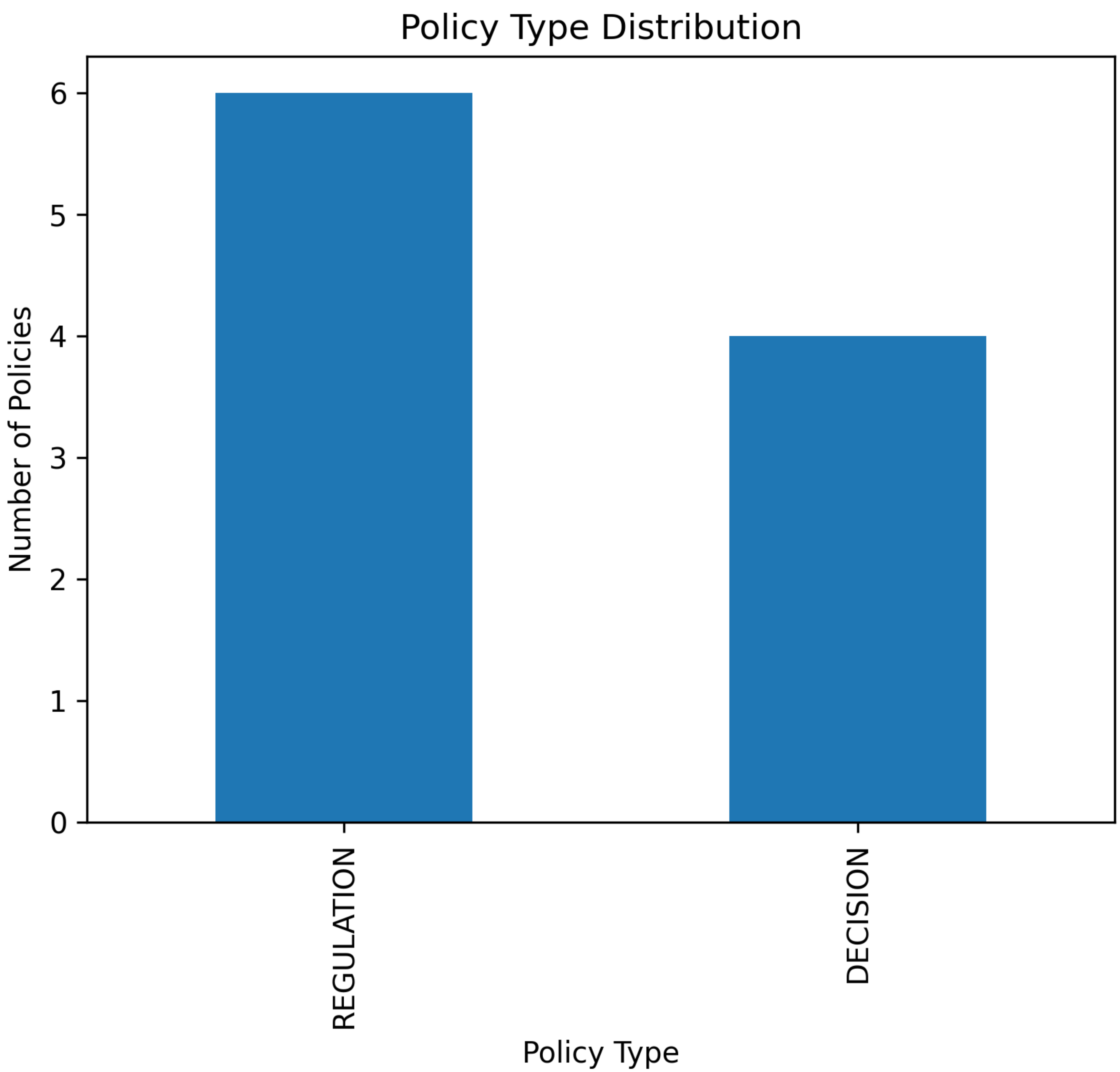
Policies by Year

Count of EU environmental/climate policies enacted per year.



Policy Type Distribution

Breakdown of policy dataset by policy type.



Policy Types by Year

Policy type composition over time.